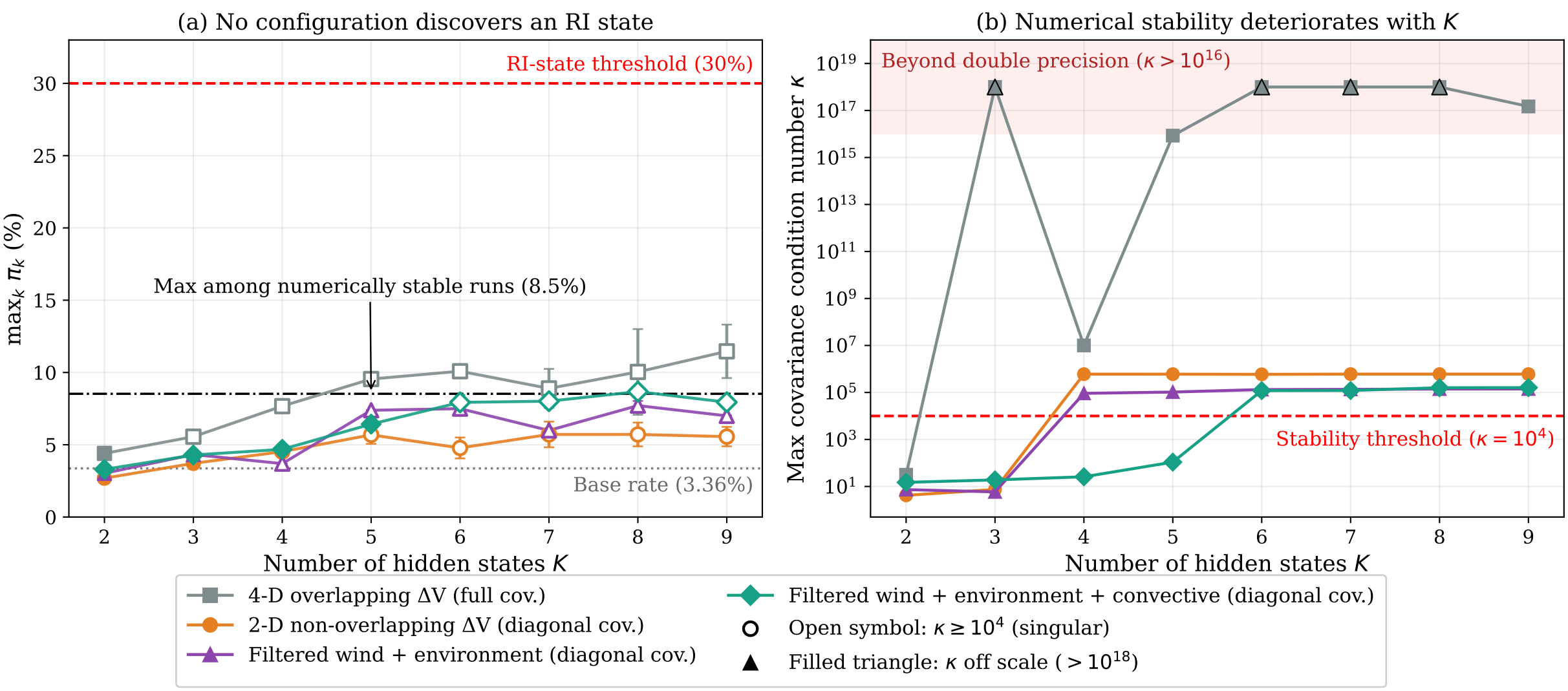
(a) No configuration discovers an RI state
RI-state threshold (30%)
Max among numerically stable runs (8.5%)
Base rate (3.36%)
$\max_k \pi_k$ (%)
Number of hidden states K
(b) Numerical stability deteriorates with K
Beyond double precision ($\kappa > 10^{16}$)
Stability threshold ($\kappa = 10^4$)
Max covariance condition number κ
Number of hidden states K
4-D overlapping ΔV (full cov.)
2-D non-overlapping ΔV (diagonal cov.)
Filtered wind + environment (diagonal cov.)
Filtered wind + environment + convective (diagonal cov.)
Open symbol: $\kappa \geq 10^4$ (singular)
Filled triangle: κ off scale ($> 10^{18}$)